eMithran Forging Costing & CAD Intelligence Architecture

High-Level (HLD) & Low-Level Design (LLD) Specification

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Document Owner	eMithran Core Manufacturing Architecture Team

1. Executive Summary & Architectural Scope

This document defines the comprehensive High-Level Design (HLD) and Low-Level Design (LLD) for the eMithran Forging Costing and Manufacturing Intelligence Module. Following the architectural paradigms established in eMithran for Sheet Metal, CNC Machining, High Pressure Die Casting (HPDC), Gravity Die Casting (GDC), and Injection Molding, the Forging Module is architected as a Domain-Driven Design (DDD) micromodule embedded within the NestJS monolith backend, coupled with physics-based geometric feature extraction in the Python OpenCASCADE CAD Engine (cad-engine), and backed by PostgreSQL with Row-Level Security (RLS).

> Core Architectural Mandate: 1. Pure Calculation Engines: Cost and physics engines perform deterministic calculations without in-flight database I/O.
> 2. Transparent Breakdown: Every line item is decomposed into Material, Machine (MHR), Labor (LHR/Wage Grade), Tooling Amortisation, Secondary Operations, and Scrap Recovery.
> 3. OpenCASCADE CAD Extraction: True B-Rep topological feature extraction directly from STEP AP203/AP214/AP242 solid models.
> 4. Dual Tenancy & Zero Data Leakage: All client CAD models and rate tables reside securely with Postgres RLS.

Module Domain	Component Location	Primary Technology	Responsibility Boundary
CAD Feature Engine	cad-engine/forging/	Python 3.10 + OpenCASCADE (PythonOCC)	Parting plane, draft angles, ribs/webs, radii, undercuts, billet sizing
Backend Costing	backend/src/modules/bom-items/costing/forging/	NestJS + TypeScript	Routing, machine matching, cycle time synthesis, tooling amortisation
Reference & Lookup	machining_reference_data & forging_lookup_tables	PostgreSQL 15 + PostgREST RLS	Flow stress curves, friction factors, flash ratios, complexity tables
Machine & Labor DB	mhr_records, lhr_records	PostgreSQL RLS	Press/hammer MHR rates, crew sizes, regional wage grade buckets (USA/IN/MX/EU)

Frontend Intelligence	components/bom/forging-intel/	Next.js 14 + React + Three.js	3D Parting line visualizer, DFM draft angle heatmaps, cost drivers breakdown

2. High-Level Design (HLD): System Architecture & Data Flow

The Forging Module sits seamlessly within eMithran's CAD-to-Cost pipeline. The architecture guarantees zero side effects, asynchronous processing for large assemblies via Redis/Postgres job queues, and sub-second calculation latency once geometric parameters are extracted.

```
flowchart TD
    User([Enterprise User / Estimator]) -->|Upload STEP / CAD| FE[Next.js Frontend]
    FE -->|POST /api/v1/cad-uploads| Gateway[NestJS Monolith API Gateway]
    Gateway -->|Queue Job| PG_Job[(PostgreSQL Queue / RLS)]
    PG_Job -->|Process STEP| CE[Python CAD Engine FastAPI]

    subgraph CAD_Engine ["CAD Engine (OpenCASCADE / cad-engine/forging)"]
        CE --> StepParser[B-Rep TopoDS_Shape Parser]
        StepParser --> PartingDetect[Parting Line & Split Plane Detector]
        PartingDetect --> Silhouette[Parting Plane Projected Area & Perimeter]
        Silhouette --> DraftClassify[Draft Angle & Undercut Classifier]
        DraftClassify --> RibWebRadius[Rib, Web & Fillet Radius Analyzer]
        RibWebRadius --> BilletCurve[Cross-Sectional Area Distribution A(x)]
        BilletCurve --> FeatGraph[Build ForgingFeatureGraphV2]
    end

    FeatGraph -->|JSON Feature Payload| Gateway
    Gateway --> ForgingService[ForgingCostService / BOMItemsService]

    subgraph Forging_Engine ["Backend Forging Cost Engine (backend/.../costing/forging)"
        ForgingService --> FeatOps[forging-feature-operations.ts: Select Operations]
        ForgingService --> RouteSelect[forging-process-choice.ts: Route Optimization]
        ForgingService --> MatResolve[forging-materials.ts: Billet Sizing & Gross Wt]
        ForgingService --> MachSelect[forging-machine.ts: Press Tonnage & Energy]
        ForgingService --> ToolCost[forging-tooling.ts: Die Block, CNC, EDM & Life]
        ForgingService --> CycleTime[forging-calculators: Physics Cycle Times]

        FeatOps --> CostSynth[forging-cost-summary.ts: Pure Cost Synthesis]
        RouteSelect --> CostSynth
        MatResolve --> CostSynth
        MachSelect --> CostSynth
        ToolCost --> CostSynth
        CycleTime --> CostSynth
    end

    CostSynth --> DB[(PostgreSQL Database)]
    DB -.->|MHR Rates| MachSelect
    DB -.->|LHR Wage Grades| MachSelect
    DB -.->|Flow Stress / Tables| CycleTime
    CostSynth -->|CostSummaryDto & DFM Report| Gateway
    Gateway -->|Interactive 3D & Cost Drivers| FE
```

The execution model enforces that: (1) CAD geometry analysis occurs once and is cached immutably by geometry hash; (2) Scenario analysis (e.g. changing factory location from USA to India or changing batch quantity from 500 to 20,000) re-runs only the pure calculation engine without re-invoking the CAD kernel.

3. CAD Feature Extraction Engine (cad-engine/forging/)

The Python OpenCASCADE feature extraction engine extracts B-Rep solid model characteristics essential for closed-die, open-die, and upset forging. It mirrors the design pattern of ‘DieCastingFeatureExtractor‘ and ‘SheetMetalFeatureExtractor‘.

3.1 Parting Line and Parting Plane Detection

Forging dies consist of an Upper Die (Hammer/Ram) and Lower Die (Anvil/Bed). The parting surface is determined by analyzing surface normals relative to candidate pressing vectors (typically the 3 principal inertia axes X, Y, Z):
- Surface Normal Classification: For each face F, calculate the dot product of face normal n with pull vector v. Faces with n • v > 0 belong to the upper die; faces with n • v < 0 belong to the lower die.
- Silhouette Perimeter: The locus of points where n • v = 0 forms candidate parting line segments.
- Optimal Split Selection: Evaluates draft angle balance, minimizes projected area overhangs, and avoids trapped undercut volumes.

3.2 Projected Area, Perimeter & Flash Gutter Footprint

The required press tonnage is directly proportional to the total projected area on the parting plane:
- Part Projected Area (Ap): The 2D polygonal shadow of the 3D solid projected onto the parting plane normal.
- Flash Land Area (Af): For closed-die forging, excess material escapes into a flash gutter. The flash land width (wf) typically ranges from 3% to 6% of the part width. Af = Lp ×wf, where Lp is the parting line perimeter.
- Total Projected Area: Atot = Ap + Af.

3.3 Draft Angle & Undercut Analysis

Draft angles are mandatory to extract the forged component from die cavities without galling or die sticking:
- External Draft Angles: Standard steel forgings require 3° to 7° (aluminum 1° to 3°).
- Internal Draft Angles (Cavity / Ribs): Sinking recesses shrink onto die projections during cooling; internal draft requires 5° to 10°.
- Undercut Detection: Faces where surface normal points away from the pull direction (re-entrant angles) are classified as Undercuts. Undercuts cannot be forged in standard two-piece dies and require either side-action wedges, split dies, or secondary machining.

3.4 Ribs, Webs, and Fillet Radii Detection

Fillet and corner radii dictate metal flow into cavities and prevent premature die cracking from thermal-mechanical fatigue:
- Corner Radius (rc): Radius on male die projections. Minimum rc = 1.5 mm to 3.0 mm + 0.05 ×cavity depth.
- Fillet Radius (rf): Radius at inside cavity corners. Sharp fillets cause cold shuts, flow folding, and stress concentrations.
- Web Thickness (tw) and Rib Aspect Ratio: The engine computes the ratio of rib height (hr) to rib width (wr). Ratios hr / wr > 4:1 trigger DFM warnings for high forging pressure and die wear.

3.5 Cross-Sectional Area Distribution Curve A(x)

To determine preforming requirements (fullering, edging, roll forging), the CAD engine slices the

solid along the principal forging axis into N equidistant planes (typically 50-100 slices), calculating cross-sectional area A(x). Regions where dA/dx exceeds critical thresholds indicate the necessity of preforming operations to avoid flash wastage and folding defects.

4. Forging Processes Taxonomy & Process Routing

eMithran classifies forging into 6 distinct manufacturing processes, each characterized by specific tooling architectures, machine kinematics, and economic batch thresholds:

Process Name	Key Characteristics	Typical Batch Size	Tolerance Class	Common Applications
Closed-Die (Impression) Hot Forging	Two-piece dies with flash gutter; heated billet (1100℃-1250℃ for steel); high strength-to-weight	500 - 100,000+	DIN 7526 Grade F / E	Automotive crankshafts, connecting rods, steering knuckles, aerospace brackets
Warm Forging	Forged between 650℃ - 850℃; reduced scale oxidation; closer net-shape tolerances	5,000 - 200,000+	DIN 7526 Grade E	Transmission bevel gears, CV joint spiders, constant velocity components
Cold Forging / Heading	Room temperature; high dimensional accuracy; work hardening; requires phosphate/soap lube	10,000 - 1,000,000+	IT8 - IT10	High-strength fasteners, spark plug shells, drive pinions, stepped shafts
Open-Die (Smith / Hammer) Forging	Flat or simple contour dies; free lateral metal flow; operator/manipulator intensive; low tooling cost	1 - 100	Rough Envelope (–5-15mm)	Heavy turbine shafts, large marine crane hooks, generator rotors, prototype blanks
Horizontal Upset Forging	Multi-stage heading & gripping dies; bar stock heated at end; accumulates volume locally	1,000 - 100,000+	DIN 7526 Grade F	Engine valves, axle shafts, flanged tubes, track pins
Seamless Ring Rolling	Radial-axial rolling mills; hollow doughnut preform rolled into high-integrity ring	50 - 10,000+	ISO 3134 Class 2	Bearing raceways, jet engine casing rings, wind turbine slewing rings, pipe flanges

4.1 Route Selection Decision Algorithm

The automated route selector (‘forging-process-choice.ts‘) compares feasible routes based on total cost per part (piece cost + die tooling per part amortization). Feasibility is evaluated against: (1) Part envelope and bounding weight; (2) Annual production volume; (3) Geometric complexity score; (4) Alloy forgeability rating.

5. Lookup Tables & Empirical Engineering Reference Data

The Forging Module integrates reference tables seeded into ‘machining_reference_data‘ and queried

through ‘forging-lookup-tables.ts‘. These tables capture empirical manufacturing constants proven across industry standards (DIN 7526, EN 10254, ASM Handbook Vol 14A).

5.1 Material Flow Stress Constants (?f)

Flow stress ?f (MPa) represents resistance to plastic deformation as a function of temperature T (°C), strain ?, and strain rate ?_dot (s?„):
?f = K • (?_dot)^m • exp(Q / RT)
Where K is strength coefficient, m is strain-rate sensitivity, and Q is activation energy.

Material Grade	Forging Temp Window (°C)	Flow Stress ?f (MPa)	Oxidation Scale Factor (%	Forgeability Index (0-10)
AISI 1020 / C20	1150 - 1250	80 - 120	2.0%	9.5 (Excellent)
AISI 1045 / C45	1100 - 1220	110 - 150	2.2%	9.0 (Very Good)
AISI 4140 / 42CrMo4	1050 - 1200	140 - 190	2.5%	8.0 (Good)
AISI 4340 / 34CrNiMo6	1050 - 1180	160 - 220	2.5%	7.5 (Moderate)
SS 304 / 1.4301	1050 - 1200	190 - 260	1.0% (Protective scale)	6.5 (Work hardens)
SS 316L / 1.4404	1050 - 1180	210 - 280	1.0%	6.0 (High hot strength)
Al 6061-T6	430 - 480	60 - 90	0.0% (Zero scale loss)	8.5 (Narrow temp band)
Al 7075-T6	380 - 440	90 - 130	0.0%	6.0 (Crack sensitive)
Ti-6Al-4V (Grade 5)	900 - 980 (Alpha-Beta)	240 - 350	0.5% (Alpha case defect)	4.0 (Narrow beta transus)
Inconel 718	980 - 1120	320 - 480	0.2%	2.5 (Severe tool wear)

5.2 Shape Complexity Factor (Cs / S1, S2)

According to DIN 7526, part complexity is classified by the ratio of part volume (Vp) to bounding circumscribed cylinder volume (Vc):
- Class S1 (Simple): Vp / Vc > 0.6. Flat, circular, or block shapes with minimal ribs (Cs = 1.1 - 1.3).
- Class S2 (Moderate): 0.3 < Vp / Vc ? 0.6. Simple flanges, shafts, discs with moderate bosses (Cs = 1.3 - 1.6).
- Class S3 (Complex): 0.15 < Vp / Vc ? 0.3. Deep ribs, thin webs, asymmetrical arms (Cs = 1.6 - 2.0).
- Class S4 (Severe / Very Complex): Vp / Vc ? 0.15. Multiple thin ribs, deep pockets, tall bosses (Cs = 2.0 - 2.5).

5.3 Flash Land Sizing and Ratios

The flash land width-to-thickness ratio (wf / tf) controls cavity backpressure ensuring complete filling before flash freezes:
- Flash thickness tf = 0.016 ×sqrt(Ap) + 0.5 mm.
- Flash land width wf = 3.0 ×tf to 5.0 ×tf.
- Flash weight typically constitutes 10% to 25% of gross billet weight.

5.4 Die Wear & Tool Life Expectancy

Die life (blow count before redressing or replacement) depends on die steel grade (DIN 1.2714 / AISI H13 / H11), forging temperature, and surface treatment:
- Uncoated H13 on Carbon Steel: 8,000 - 15,000 parts.
- Gas Nitrided / Plasma Nitrided H13: 20,000 - 35,000 parts.
- Hardfaced Stellite / Maraging Inlay: 40,000 - 70,000 parts.
- Stainless Steel Forging: 0.6×steel die life; Titanium Forging: 0.4×steel die life.

6. Machine Database & Selection Engine (MHR & Press Kinematics)

Forging machinery is stored in ‘mhr_records‘ and queried through ‘forging-machine.ts‘. Presses and hammers differ fundamentally in their energy and force delivery characteristics:

6.1 Machine Classes & Kinematics

Machine Class	Force / Energy Delivery	Kinematics & Contact Time	Primary Application
Hydraulic Forging Press	Constant rated force throughout stroke; 500T - 30,000T	Slow ram speed (10-50 mm/s); long die contact; high heat transfer	Large forgings, isothermal forging, heavy open-die, titanium & nickel
Mechanical Crank Press	Displacement-dependent force (peaks near Bottom Dead Center); 300T - 8,000T	Rapid sinusoidal motion (30-90 SPM); short contact time; high productivity	High-volume closed die automotive, connecting rods, gears
Screw Press (Direct Drive / Friction)	Fixed kinetic energy per blow; 400T - 6,300T; self-limiting load	Intermediate speed (100-300 mm/s); bounce-back stroke; high accuracy	Turbine blades, precision bevel gears, hand tools
Drop Hammer (Air / Steam / Counterblow)	High kinetic energy E = 0.5•m•v†; strain rates up to 100 s?„; 10 kJ - 500 kJ	Impact duration < 10 ms; minimal die chilling; multi-blow progressive shaping	Medium to heavy steel forgings, complex shapes with multi-impression dies
Horizontal Upsetter	Horizontal heading slide + vertical gripping slide; 2" to 10" bar capacity	Multi-cavity sequence; automatic bar feed	Long shafts with upsets, valves, flanged axles
Radial-Axial Ring Rolling Mill	Continuous radial and axial rolling forces; OD 200 mm - 5000 mm	Continuous rotary deformation; fast cycle time (20-90 s)	Seamless rings, bearing races, aerospace flanges

6.2 Physics-Based Tonnage & Energy Formulation

The required press forging load F_req (kN) is determined via the slab method formulation:
F_req = ?f • Cs • Atot • [1 + (2 • ? • r_avg) / (3 • h_avg)] / 1000
Where:
- ?f = Flow stress of alloy at forging finish temperature (MPa)
- Cs = Shape complexity factor (1.1 to 2.5)
- Atot = Total projected area including flash land (mm†)
- ? = Friction coefficient at die-billet interface (0.25 cold, 0.40 hot with graphite)
- r_avg = Characteristic radius of forging = sqrt(Ap / ?) (mm)
- h_avg = Mean thickness of forging = Volume / Ap (mm)

For hammers and screw presses, required blow energy E_req (kJ) is calculated as:
E_req = 1.35 • Vp • ?f • ln(h0 / h1) / 1000

Where h0 is initial billet height and h1 is final forged height.

6.3 Machine Hour Rate (MHR) Breakdown Model

MHR represents the hourly cost of operating the machine (excluding direct operating labor crew, which is costed separately via LHR). The formula enforces financial rigor:
MHR ($/hr) = CapEx_Depreciation + Electricity_Cost + Gas_Furnace_Cost + Die_Bolster_Floor_Space + Maintenance_Spares + Lubrication_Pumping
Where:
- CapEx Depreciation = (Machine Purchase Price - Residual Value) / (Useful Life Years ×Annual Operating Hours ×OEE)
- Electricity Cost = Connected Power (kW) ×Utilization Factor ×Industrial Power Rate ($/kWh)
- Maintenance & Wear = 8% to 12% of annual capital expenditure.

7. Raw Materials, Billet Sizing & Yield Economics

Material cost represents 35% to 60% of total forging piece cost. The raw material engine computes exact billet dimensions, gross input weight, and net weight from 3D CAD density.

7.1 Gross Weight Synthesis Formula

Gross Billet Weight (W_gross) is synthesized from net part weight and all manufacturing yield losses:
W_gross = W_net + W_flash + W_scale + W_tong + W_kerf
1. Net Part Weight (W_net): Volume (Vp from CAD) ×Density (? from ‘raw_materials‘).
2. Flash Loss (W_flash): Total flash land volume ×Density. Typically 12% - 25% of W_net.
3. Scale Oxidation Loss (W_scale): Hot steel forms iron oxide scale in gas furnaces (2.0% - 3.0% of heated weight). In induction furnaces with inert atmosphere, scale loss drops to 0.5% - 1.0%.
4. Tong Hold Loss (W_tong): Gripping margin on billet bar (required in open-die and hammer forgings; zero in automated transfer presses).
5. Saw Kerf Loss (W_kerf): Material converted to chips during billet cutoff = (?/4 • D† • saw_blade_thickness) ×?.

7.2 Scrap Recovery & Buyback Valuation

Scrap generated from trimming flash and saw kerf is credited back at commodity scrap buyback rates:
Net Material Cost = (W_gross ×Raw_Material_Cost_Per_Kg) - (W_scrap ×Scrap_Credit_Rate_Per_Kg)
Where W_scrap = W_flash + W_kerf + W_tong (scale loss cannot be credited as solid scrap).

8. Wage Grade & Labor Rate Architecture (LHR)

Forging is a crew-based manufacturing operation. Unlike solitary CNC machining, a heavy forging press line operates with a synchronized crew consisting of distinct skill tiers.

8.1 Forging Crew Hierarchy & Wage Grade Categories

Wage Grade / Role	Skill Category	Key Operational Responsibi	Typical Line Allocation
Master Forger / Head Press Lead	Grade A+ (Specialist)	Press setup, die alignment, stroke tuning, first-piece inspection, closed-die balance	1 per forging line (leads crew)

Press Operator	Grade A (Skilled)	Operating press controls, lubricating dies with spray wand/auto-sprayer	1 per press
Manipulator / Crane Driver	Grade A (Skilled)	Guiding heavy billets with hydraulic manipulator into hammer/press dies	1 for open-die & heavy presses (>2500T)
Billet Heater / Furnace Loader	Grade B (Semi-Skilled)	Monitoring induction heater / gas furnace temperature, pyrometer verification, descaling transfer	1 per line
Trimming & Piercing Operator	Grade B (Semi-Skilled)	Transferring hot forging to trim press, punching out flash and web plugs	1 per trim press
Heat Treatment Technician	Grade A (Skilled)	Furnace cycle management, quench tank agitation, hardness testing (Brinell/Rockwell)	Shared across heat treat department (0.25 FTE/part)
Fettling / Grinding / Blasting Labor	Grade C (General)	Operating shot blast tumbler, manual grinding of trim remnants / parting fins	1 per finishing station
NDT & Quality Inspector	Grade A+ (Certified Level II)	Magnetic Particle Inspection (MPI), Ultrasonic flaw testing, CMM dimensional verification	1 per quality batch cell

8.2 Digital Factory Regional Wage Benchmarks

eMithran benchmarks Labor Hour Rates (LHR) across its 6 global manufacturing regions (stored in ‘lhr_records‘ and updated quarterly). Forging operations scale labor costs based on regional crewed rates:

- United States (Midwest): Grade A+ = $65/hr | Grade A = $48/hr | Grade B = $34/hr | Grade C = $26/hr
- Mexico (Monterrey / Bajio): Grade A+ = $16/hr | Grade A = $11/hr | Grade B = $7.50/hr | Grade C = $5.20/hr
- India (Pune / Chennai): Grade A+ = $9.50/hr | Grade A = $6.20/hr | Grade B = $4.10/hr | Grade C = $2.80/hr
- Poland / CEE: Grade A+ = $28/hr | Grade A = $20/hr | Grade B = $14/hr | Grade C = $10.50/hr
- Germany (Baden-Württemberg): Grade A+ = $82/hr | Grade A = $62/hr | Grade B = $45/hr | Grade C = $34/hr
- Vietnam (Hai Phong): Grade A+ = $8.00/hr | Grade A = $5.20/hr | Grade B = $3.50/hr | Grade C = $2.40/hr

9. Operations Taxonomy & Standard Routing Workflow

Forging operations are registered in ‘process_taxonomy_operations‘. A complete standard manufacturing route consists of 13 sequential operations (OP 10 to OP 130):

Op Nbr	Operation Name	Machine Class	Cycle Time Formulation	Labor Crew Allocation
OP 10	Billet Cutoff	Automatic Bandsaw / Cold Billet Shear	T_cut = Area_bar / Cut_rate (mm†/min) + 5s index	1 Operator per 3 saws (0.33 FTE)
OP 20	Billet Preheating	Medium-Frequency Induction Furnace / Gas Furnace	T_heat = Continuous feed; cycle matched to press stroke (e.g. 15-45s)	1 Furnace Loader (Grade B)
OP 30	Billet Descaling	High-Pressure Waterjet (200-300 bar) / Scraping Rolls	T_desc = 2.0 to 4.0 seconds per slug	Automated / In-line conveyor
OP 40	Preforming / Busting	Fullering / Edging Die Cavity or Cross-Wedge Rolls	T_preform = 1 to 2 blows / strokes (4-8s)	Included in main forge press crew
OP 50	Blocker Impression Forging	Forging Press / Hammer (Blocker Cavity)	T_block = 1 stroke (press) or 2-3 blows (hammer) (6-12s)	Main Forge Press Crew (Lead + Operator)
OP 60	Finisher Impression Forging	Forging Press / Hammer (Finisher Cavity)	T_finish = 1 stroke or 2 blows + die blow/lube (8-16s)	Main Forge Press Crew
OP 70	Hot Trimming & Piercing	Mechanical Crank Trim Press (100T - 600T)	T_trim = 1 single stroke + unload flash (4-8s)	1 Trimming Operator (Grade B)
OP 80	Coining / Sizing (Restriking)	Heavy Knuckle-Joint Coining Press	T_coin = 1 stroke (optional, for tight tolerances) (5-8s)	1 Coining Press Operator (Grade B)
OP 90	Heat Treatment	Continuous Roller Hearth / Batch Car-Bottom Furnace	T_ht = Batch cycle (Normalizing, Q&T, Isothermal Anneal)	Shared Furnace Specialist (0.2 FTE)
OP 100	Surface Descaling & Shot Blast	Rotary Barrel / Monorail Hanger Shot Blaster	T_blast = Batch cycle (8-15 min per basket / 50 parts)	1 Finishing Technician (Grade C)
OP 110	Secondary Machining	CNC Horizontal / Vertical Machining Center	Calculated via Machining Module (milling, drilling, tapping)	1 CNC Machinist per 2 machines (0.5 FTE)
OP 120	Inspection & NDT	Magnetic Particle (MPI) / Ultrasonic (UT) / CMM	T_insp = Statistical sampling (AQL) or 100% safety critical (30-120s)	1 Certified Inspector Level II (Grade A+)
OP 130	Tooling & Die Amortisation	Die Maintenance Shop / CNC Sinker EDM	Amortised = Total_Die_Set_Cost / Total_Production_Volume	Die Shop Amortised Capital

9.1 Die Tooling Cost Model

Forging die tooling represents a substantial non-recurring engineering (NRE) investment. Total die cost comprises:

Die_Cost = Die_Steel_Material + CNC_Rough_Machining + CNC_Finish_Machining + EDM_Die_Sinking +

Heat_Treatment_Die + Die_Polishing + Bolster_Mounting
Where:
- Die Block Material = Volume of Upper + Lower Die Blocks ×Density ×Steel Cost/kg (DIN 1.2714 / AISI H13).
- Die cavity volume is sized to ensure 3×to 5×part depth thickness to resist bursting pressures.
- Per-piece tooling cost = Die_Cost / N_pieces_total (or amortised per batch if replaced).

10. Low-Level Design (LLD): Code Modules, Interfaces & SQL Schemas

This section specifies the exact TypeScript interfaces, Python classes, and PostgreSQL schema definitions required to implement the Forging Module across the repository.

10.1 Backend TypeScript Architecture (backend/src/modules/bom-items/costing/forging/)

```
// backend/src/modules/bom-items/costing/forging/forging-types.ts

export type ForgingProcessType =
  | 'CLOSED_DIE_HOT'
  | 'WARM_FORGING'
  | 'COLD_HEADING'
  | 'OPEN_DIE_HAMMER'
  | 'UPSET_FORGING'
  | 'SEAMLESS_RING_ROLLING';

export interface ForgingGeometryInput {
  netVolumeMm3: number;
  projectedAreaMm2: number;
  partingPerimeterMm: number;
  partingPlaneNormal: [number, number, number];
  boundingBoxDimsMm: [number, number, number];
  maxRibHeightMm: number;
  minWebThicknessMm: number;
  minFilletRadiusMm: number;
  minCornerRadiusMm: number;
  undercutVolumeMm3: number;
  hasTrappedUndercuts: boolean;
  crossSectionAreaCurve: Array<{ positionMm: number; areaMm2: number }>;
}

export interface ForgingMaterialSpecs {
  grade: string;
  family: string;
  densityKgM3: number;
  forgingTempStartC: number;
  forgingTempFinishC: number;
  flowStressMpa: number;
  scaleLossPct: number;
  costPerKg: number;
  scrapCreditPerKg: number;
}

export interface ForgingMachineSpec {
  machineId: string;
  name: string;
  machineClass: 'HYDRAULIC_PRESS' | 'MECHANICAL_PRESS' | 'SCREW_PRESS' | 'DROP_HAMMER' |
  ratedTonnageKn: number;
  energyCapacityKj?: number;
  bedWidthMm: number;
  bedDepthMm: number;
  maxStrokeMm: number;
```

```
  strokesPerMinute: number;
  machineRatePerHr: number;
  requiredCrewSize: number;
  crewWageGradeDistribution: Record<string, number>; // e.g. { 'GRADE_A_PLUS': 1, 'GRADE
}

export interface ForgingCostSummaryDto {
  partId: string;
  process: ForgingProcessType;
  batchSize: number;
  annualVolume: number;

  // Weights (kg)
  netWeightKg: number;
  flashWeightKg: number;
  scaleLossKg: number;
  grossBilletWeightKg: number;
  materialYieldPct: number;

  // Cost Breakdown ($)
  rawMaterialCost: number;
  scrapCredit: number;
  netMaterialCost: number;

  forgingMachineCost: number;
  forgingLaborCost: number;
  heatingCost: number;
  trimmingCost: number;
  heatTreatmentCost: number;
  shotBlastingCost: number;
  secondaryMachiningCost: number;
  inspectionNdtCost: number;

  totalProcessCost: number;
  dieToolingSetCost: number;
  dieToolingAmortisedPerPart: number;

  piecePartCost: number;
  totalPieceCostWithTooling: number;

  // DFM Warnings
  dfmWarnings: Array<{ code: string; severity: 'INFO' | 'WARNING' | 'BLOCKER'; message:
}
```

10.2 Pure Calculation Engine (forging-cost-engine.ts)

```
// backend/src/modules/bom-items/costing/forging/forging-cost-engine.ts

import { ForgingGeometryInput, ForgingMaterialSpecs, ForgingMachineSpec, ForgingCostSumm

export function calculateRequiredForgingTonnageKn(
  geom: ForgingGeometryInput,
  mat: ForgingMaterialSpecs,
  complexityFactor: number = 1.4,
  frictionCoeff: number = 0.35,
): number {
  // Flash land sizing
  const flashThicknessMm = 0.016 * Math.sqrt(geom.projectedAreaMm2) + 0.5;
  const flashLandWidthMm = 3.5 * flashThicknessMm;
  const flashAreaMm2 = geom.partingPerimeterMm * flashLandWidthMm;
  const totalProjectedAreaMm2 = geom.projectedAreaMm2 + flashAreaMm2;
```

```
  const charRadiusMm = Math.sqrt(geom.projectedAreaMm2 / Math.PI);
  const avgThicknessMm = Math.max(2.0, geom.netVolumeMm3 / geom.projectedAreaMm2);

  // Slab method with shape complexity and die friction
  const pressureMultiplier = 1.0 + (2.0 * frictionCoeff * charRadiusMm) / (3.0 * avgThic
  const requiredLoadN = mat.flowStressMpa * complexityFactor * totalProjectedAreaMm2 * p

  return Math.round(requiredLoadN / 1000.0);
}

export function synthesizeGrossBilletWeightKg(
  geom: ForgingGeometryInput,
  mat: ForgingMaterialSpecs,
): { netWeightKg: number; flashWeightKg: number; scaleLossKg: number; grossWeightKg: num
  const netWeightKg = (geom.netVolumeMm3 * 1e-9) * mat.densityKgM3;

  const flashThicknessMm = 0.016 * Math.sqrt(geom.projectedAreaMm2) + 0.5;
  const flashLandWidthMm = 3.5 * flashThicknessMm;
  const flashVolumeMm3 = geom.partingPerimeterMm * flashLandWidthMm * flashThicknessMm;
  const flashWeightKg = (flashVolumeMm3 * 1e-9) * mat.densityKgM3;

  const subTotalKg = netWeightKg + flashWeightKg;
  const scaleLossKg = subTotalKg * (mat.scaleLossPct / 100.0);
  const sawKerfLossKg = 0.05 * netWeightKg; // Standard 5% saw / crop end allowance

  const grossWeightKg = subTotalKg + scaleLossKg + sawKerfLossKg;
  return { netWeightKg, flashWeightKg, scaleLossKg, grossWeightKg };
}
```

10.3 Python OpenCASCADE Feature Extractor (cad-engine/forging/)

```
# cad-engine/forging/feature_extractor.py
from __future__ import annotations
import math
import logging
from typing import Any, Dict, List, Optional
from OCC.Core.BRepGProp import brepgprop
from OCC.Core.GProp import GProp_GProps
from OCC.Core.TopAbs import TopAbs_FACE
from OCC.Core.TopExp import TopExp_Explorer
from OCC.Core.BRepAdaptor import BRepAdaptor_Surface
from OCC.Core.GeomAbs import GeomAbs_Plane, GeomAbs_Cylinder

logger = logging.getLogger(__name__)

class ForgingFeatureExtractor:
    """Extracts B-Rep geometric features required for forging costing and DFM."""

    def extract(self, shape: Any, bbox_dims: List[float], pull_axis: Optional[List[float
        axis = pull_axis if pull_axis else [0.0, 0.0, 1.0]

        props = GProp_GProps()
        brepgprop.VolumeProperties(shape, props)
        net_volume = props.Mass()

        brepgprop.SurfaceProperties(shape, props)
        surface_area = props.Mass()

        # Analyze draft angles on all faces
        exp = TopExp_Explorer(shape, TopAbs_FACE)
        draft_angles: List[float] = []
        undercuts: List[Dict[str, Any]] = []
```

```
        parting_line_faces: List[int] = []
        face_idx = 0

        while exp.More():
            face = exp.Current()
            surf = BRepAdaptor_Surface(face)
            # Evaluate midpoint normal
            u_mid = (surf.FirstUParameter() + surf.LastUParameter()) / 2.0
            v_mid = (surf.FirstVParameter() + surf.LastVParameter()) / 2.0
            pnt, d1u, d1v = surf.D1(u_mid, v_mid)
            normal = d1u.Crossed(d1v)
            if normal.Magnitude() > 1e-6:
                normal.Normalize()
                cos_theta = normal.X() * axis[0] + normal.Y() * axis[1] + normal.Z() * a
                angle_deg = 90.0 - math.degrees(math.acos(max(-1.0, min(1.0, cos_theta))
                draft_angles.append(angle_deg)
                if abs(angle_deg) < 1.0:
                    parting_line_faces.append(face_idx)
                elif angle_deg < -0.5:
                    undercuts.append({"face_id": face_idx, "angle_deg": round(angle_deg,
            face_idx += 1
            exp.Next()

        # Compute silhouette projected area on parting plane
        projected_area = bbox_dims[0] * bbox_dims[1] * 0.785 # Bounding ellipse approxim
        parting_perimeter = 2.0 * (bbox_dims[0] + bbox_dims[1]) * 1.15

        return {
            "domain": "forging",
            "net_volume_mm3": round(net_volume, 2),
            "surface_area_mm2": round(surface_area, 2),
            "projected_area_mm2": round(projected_area, 2),
            "parting_perimeter_mm": round(parting_perimeter, 2),
            "primary_pull_axis": axis,
            "min_draft_angle_deg": round(min(draft_angles), 2) if draft_angles else 0.0,
            "has_undercuts": len(undercuts) > 0,
            "undercut_count": len(undercuts),
            "undercuts": undercuts,
            "parting_line_face_count": len(parting_line_faces)
        }
```

10.4 Database Schemas & Migration Script (PostgreSQL)

```
-- backend/src/database/migrations/950_create_forging_module_tables.sql
-- Enables Forging Machine, Materials, and Empirical Lookup Tables with Tenancy RLS

CREATE TABLE IF NOT EXISTS public.forging_machines (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    organization_id UUID REFERENCES public.organizations(id) ON DELETE CASCADE,
    machine_name VARCHAR(100) NOT NULL,
    machine_class VARCHAR(50) NOT NULL, -- 'HYDRAULIC_PRESS', 'MECHANICAL_PRESS', 'SCREW
    rated_tonnage_kn NUMERIC(12,2) NOT NULL,
    energy_capacity_kj NUMERIC(10,2),
    bed_width_mm NUMERIC(10,2) NOT NULL,
    bed_depth_mm NUMERIC(10,2) NOT NULL,
    max_stroke_mm NUMERIC(10,2) NOT NULL,
    strokes_per_minute NUMERIC(6,2) NOT NULL,
    machine_rate_per_hr NUMERIC(10,2) NOT NULL,
    required_crew_size INT NOT NULL DEFAULT 2,
    crew_wage_distribution JSONB NOT NULL DEFAULT '{"GRADE_A": 1, "GRADE_B": 1}',
    location VARCHAR(50) NOT NULL DEFAULT 'USA',
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
```

```
);

CREATE TABLE IF NOT EXISTS public.forging_lookup_tables (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    table_name VARCHAR(100) NOT NULL, -- 'FLOW_STRESS', 'FRICTION_FACTOR', 'COMPLEXITY_F
    alloy_grade VARCHAR(50),
    key VARCHAR(100) NOT NULL,
    numeric_value NUMERIC(14,4),
    json_data JSONB,
    source_standard VARCHAR(100) DEFAULT 'DIN_7526 / ASM_VOL_14A',
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

-- Enable Row Level Security (RLS)
ALTER TABLE public.forging_machines ENABLE ROW LEVEL SECURITY;
ALTER TABLE public.forging_lookup_tables ENABLE ROW LEVEL SECURITY;

CREATE POLICY forging_machines_read_policy ON public.forging_machines
    FOR SELECT USING (organization_id IS NULL OR organization_id = auth.current_organiza

CREATE POLICY forging_lookup_tables_read_policy ON public.forging_lookup_tables
    FOR SELECT USING (true); -- Master lookup reference data is readable by all authenti
```

11. DFM Rules, Quality Assurance & Engineering Standards

The Forging Module includes automated Design for Manufacturability (DFM) verification rules that inspect geometry against industry thresholds before costing is finalized:

Rule Code	DFM Inspection Check	Threshold Condition	Severity	Engineering Recommendat
DFM-FRG-001	Zero / Insufficient External Draft	Draft Angle < 3.0° on non-parting faces	WARNING	Increase external draft to minimum 5° (steel) or 3° (aluminum) to avoid die sticking.
DFM-FRG-002	Trapped Undercut Geometry	Surface normal dot product < -0.05	BLOCKER	Cannot forge in standard 2-piece dies. Requires split die, side-action core, or secondary CNC milling.
DFM-FRG-003	Sharp Fillet Radius	Fillet Radius rf < 2.0 mm	WARNING	Risk of die cracking and cold shut flow defect. Increase fillet radius to minimum 3.5 mm.
DFM-FRG-004	Excessive Rib Aspect Ratio	Rib Height / Rib Width > 4.5:1	WARNING	High cavity pressure and rapid die wear. Taper rib walls or increase web thickness.
DFM-FRG-005	Thin Web Section	Web Thickness < 3.0 mm (Steel) / 2.0 mm (Al)	BLOCKER	Thin webs cool prematurely causing unfill defects and die damage. Thicken web.

DFM-FRG-006	High Tonnage Requirement	Required Press Tonnage > Maximum Plant Machine	BLOCKER	Part exceeds plant press capacity. Reroute to hydraulic multi-action press or open-die preform.

12. Implementation Plan & Delivery Milestones

The module implementation is partitioned into four phased engineering sprints across the 3 core layers:

Sprint Phase	Scope & Deliverables	Target Components	Acceptance Criteria
Phase 1: CAD Engine	OpenCASCADE feature extractor, parting line & draft angle detection	cad-engine/forging/	Pass 25 unit tests on automotive crankshaft and connecting rod STEP models.
Phase 2: Database & Seed	Postgres migrations 950-953, seed 45 press machines & DIN lookup tables	backend/src/database/migrations/	All tables queryable with RLS tenant isolation verified in pgTAP tests.
Phase 3: Cost Engine	Pure calculation engine, physics tonnage, gross weight, multi-wage LHR	backend/.../costing/forging/	Cost calculation latency < 15ms; accurate within –3% of benchmark quote data.
Phase 4: Frontend & DFM	Three.js 3D parting line viewer, DFM rule violation list, cost drivers	components/bom/forging-intel/	Visual display of parting line, draft angle heatmap, and one-click PDF quote export.